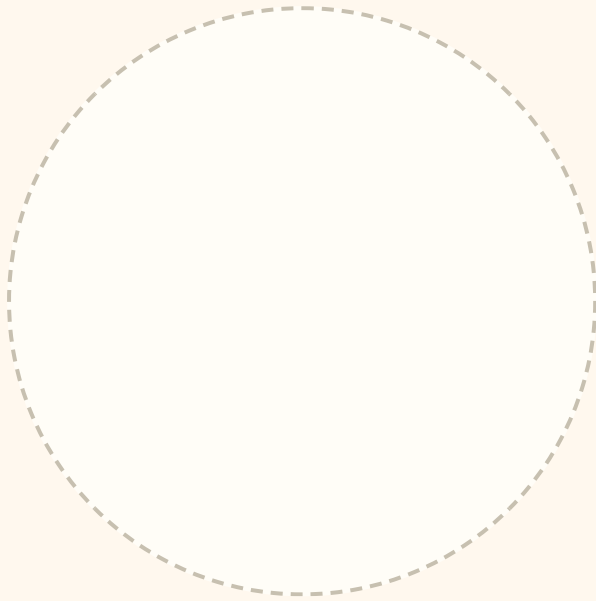


Design a world that could hold life

Invent a planet or a moon. Draw it, name it, then show how it supplies each of the five things every living thing needs.



The name of my world

The star it goes around

Where it sits

☐

too hot

☐

just right

☐

too cold

The five requirements

Write a line or two for each. Guessing is allowed. Inventing is encouraged.



Liquid water Where is it, and what keeps it from freezing or boiling?



An energy source Sunlight, or heat and chemistry from inside?



The right atoms Carbon, hydrogen, nitrogen, oxygen, phosphorus, sulphur



Some shelter Air, ice or rock between your life and space?



A lot of time Has it stayed steady for billions of years?

Now check your world

Answer honestly. These are close to the questions astronomers ask about a real planet.

1	Does my world have liquid water somewhere, even under ice?	☐ YES	☐ NO
2	Does it have an energy source that does not run out?	☐ YES	☐ NO
3	Is it big enough for its gravity to hold on to an atmosphere?	☐ YES	☐ NO
4	Is there something to block the radiation from space?	☐ YES	☐ NO
5	Has it stayed steady for a very long time?	☐ YES	☐ NO
6	Could something live there without wearing a spacesuit?	☐ YES	☐ NO

What the answers mean

6 yes Your world is a strong candidate. Astronomers would want a closer look.

4 or 5 yes Promising. Life might survive in one sheltered corner of it.

3 or fewer Harsh, but not hopeless. Remember Europa: an ocean under ice, and no sunlight at all.

One thing I would change about my world
